Task 5 — InfoNCE Mathematical Derivations

Vision Meets Language: Multimodal Transformer from Scratch | Seasons of Code (IIT Bombay)

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1. InfoNCE Loss Explicit Summation & Derivation from First Principles

Goal: Derive the image-to-text InfoNCE loss starting from the principle of maximizing the log-probability of the matching caption under a softmax distribution over candidate captions in a batch of size N.

Let s_{ij} denote the cosine similarity between normalized image embedding $\mathbf{I}_i$ and text embedding $\mathbf{T}_j$, and let $\tau > 0$ be the scalar temperature hyperparameter.
Under the categorical softmax distribution, the conditional probability that text caption j matches image query i is:

Softmax Probability:
$$p(T_j \mid I_i) = \exp(s_{ij} / \tau) / \Sigma_{k=1..N} \exp(s_{ik} / \tau)$$

For candidate i, the ground-truth matching caption is $j = i$. The log-likelihood of the correct pair is:

Log-Likelihood:
$$\log p(T_i \mid I_i) = (s_{ii} / \tau) - \log \Sigma_{k=1..N} \exp(s_{ik} / \tau)$$

The image-to-text InfoNCE loss $\text{\&Lscript;}_{i2t}$ is defined as the negative expected log-likelihood across all N batch items:

Explicit Summation Formula:
$$\text{\&Lscript;}_{i2t} = - (1 / N) \Sigma_{i=1..N} \log p(T_i \mid I_i)$$
$$\text{\&Lscript;}_{i2t} = (1 / N) \Sigma_{i=1..N} [- (s_{ii} / \tau) + \log \Sigma_{k=1..N} \exp(s_{ik} / \tau)]$$

2. Temperature Limiting Behavior ($\tau \to 0$ and $\tau \to \infty$)

Case A: Limit as $\tau \to 0$ (Zero Temperature / Sharp Softmax)

As τ approaches 0, the exponential term with the largest similarity $s_{i,max} = max_k\ s_{ik}$ completely dominates the log-sum-exp term:
$\tau \cdot \log \Sigma_{k=1..N} \exp(s_{ik} / \tau) \to \max_k s_{ik}$.
Multiplying $\text{\&Lscript;}_{i2t}$ by τ yields:
$\lim_{\tau \to 0} \tau \cdot \text{\&Lscript;}_{i2t} = (1 / N) \Sigma_{i=1..N} (\max_k s_{ik} - s_{ii})$.
To achieve zero loss, we require $max_k\ s_{ik} = s_{ii}$, implying $s_{ii} > s_{ik}$ for all $k \neq i$. This acts like a hard max-margin constraint: the correct pair similarity must strictly exceed all negative candidate similarities.

Case B: Limit as $\tau \to \infty$ (Infinite Temperature / Flat Softmax)

As $\tau \to \infty$, all scaled similarities $s_{ik} / \tau \to 0$. Therefore $\exp(s_{ik} / \tau) \to 1$ for all k.
The log-sum term becomes $\log(\Sigma_{k=1..N} 1) = \log(N)$.
Substituting this back into the loss:
$\lim_{\tau \to \infty} \text{\&Lscript;}_{i2t} = - (1 / N) \Sigma_{i=1..N} (0 - \log N) = \mathbf{\log N}$.
Intuition: At infinite temperature, the softmax distribution flattens into a uniform distribution over N classes where $p = 1/N$. The model retains zero discriminative power, collapsing the loss to maximum uncertainty log(N).

3. Analytical Gradients & Physical Interpretation of Signs

Let $P_{ij} = \exp(s_{ij} / \tau) / \Sigma_{k=1..N} \exp(s_{ik} / \tau)$ denote the softmax weight assigned to pair (i, j).

Gradient w.r.t Correct Pair Similarity (s_{ii}):
$$\partial \text{\&Lscript;}_{i2t} / \partial s_{ii} = (1 / N \tau) \cdot (P_{ii} - 1) \ \leq\ 0$$

Gradient w.r.t Incorrect Pair Similarity (s_{ij} for $j \neq i$):
∂&Lscript;$_{i2t}$ / ∂s_{ij} = (1 / N τ) · P_{ij} > 0

Interpretation of Gradient Signs:
• **Correct Pair (s_{ii}):** The gradient ∂&Lscript; / ∂s_{ii} is strictly negative (since $P_{ii} \leq 1$). Under gradient descent ($s_{ii} \leftarrow s_{ii} - \eta \nabla$), subtracting a negative quantity **increases s_{ii}**. This pulls matching image and text embeddings closer together.
• **Incorrect Pair (s_{ij}):** The gradient ∂&Lscript; / ∂s_{ij} is strictly positive (since $P_{ij} > 0$). Under gradient descent, subtracting a positive gradient **decreases s_{ij}**. This pushes mismatched image and text embeddings apart in the joint embedding space.